Maths for economists - An introductory guide

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Preface

This compendium is written to work as a complement for students at the introductory/intermediate level in economics (primarily Economics B100). The focus is on elementary mathematical tools and methods for economic analysis. The style is informal and the intention is not to be stringent and strict from a mathematicians point of view. Rather, focus is on intuition and basic understanding to facilitate learning within introductory courses in economics. In addition, the compendium is supposed to be dynamic in the sense that the content may/will change over time depending on current demand. Furthermore, existing exercise sets are supposed to expand over time with inspiration from students. Accordingly, I look forward to comments, suggestions and all other types of contributions that may help improve the text. The material varies from elementary topics such as algebra, equations, notations etc., to more sophisticated topics such as e.g. optimization of multivariate functions and matrix algebra. The compendium is partly made from my private lecture notes based on the book "Essential mathematics for economic analysis", 4th ed., by K. Sydsæter, P. Hammond and A. Strøm.

Lars Persson, Umeå 2012

1 Introduction and Foundations

Mathematics play a very central and important role in economics since theories, economic activity and human behavior to some extent can be described by 'mathematical models'. Of course, a mathematical model cannot fully predict the trends and changes in real world economies since human behavior, by nature, is more or less unpredictable. The idea with maths in economics is rather to, with help from a pre-specified model, describe and reflect important mechanisms in an economy. For example, we may define a theoretical model (based on empirical findings) for how a reduced interest rate increases investments, although it is hard to argue that you have the exact answer on how much it will be in reality. Still, given the pre-specified mathematical model you may derive how investment changes from a, say, 0.5 percentage point decrease in the interest rate. In addition to the direct use of math in theoretical models, math is fundamental for analyzing and interpreting empirical data.

1.1 Numbers

You all know that there exists something called 'numbers'. The most fundamental are 'natural numbers' (1, 2, 3, 4, ...). Among these (positive) integers there are both even numbers $(2, 4, 6, 8, ...)$ and odd numbers $(1, 3, 5, 7, ...)$. By including 0 and the negative integers $(-1, -2, -3, ...)$, all possible integers are defined. Furthermore, two integers divided by each other make what is called 'rational numbers'. In other words, examples of rational numbers could be

$\frac{1}{4}, \quad \frac{5}{7}, \quad \frac{211}{12}, \quad \frac{3}{1}, \quad \frac{0}{4} = 0, \quad -\frac{1}{4}.$

That is, rational numbers are defined so that you may write a/b, where a

and b are integers.[1] Notice in the example above that 0 and negative integers are also defined as rational numbers.

1.1.1 Addition, subtraction, multiplication and division with positive and negative numbers

My guess is that all of you are well aware how we add and subtract *positive* numbers. For example, start with 5kr and then add 10kr means that you end up with a *sum* of $5 + 10 = 15$kr. Subtracting a positive number is also very straightforward, start with 10kr and give away 3kr and you end up with $10-3 = 7$kr. However, the addition and subtraction of *negative* numbers are perhaps not equally straightforward. Think about a case where we want to add two bank accounts, one with a positive balance (+300kr) and one with a negative balance (−220kr), which can be expressed as $(+)300 + (-)220 = +80$kr. Perhaps obvious, but you now see that adding a negative number is equivalent to subtracting the 'same number', i.e. $+(-)$ becomes 'minus'. Finally, what about subtracting a negative number? Think about comparing two bank accounts by calculating the difference between them. Consider an example where one account is +400kr, while the other is −250kr (a deficit). The difference can then be expressed as $(+)400 - (-)250 = 650$kr, which means that the difference in balances is $400 + 250 = 650$kr.

The implication from the examples above is thus that

$$\begin{array}{lrl} +(+) & \text{means plus} & (+) \\ +(-), -(+) & \text{means minus} & (-) \\ -(-) & \text{means plus} & (+) \end{array}$$

[1] By switching to letters instead of numbers we are actually talking about an algebraic expression. More on algebra below.

Let us now turn to to multiplication with positive and negative numbers. A number that is multiplied by +1 remains the same, while multiplying by -1 changes its sign (a positive number becomes negative and vice versa). For example, $7 \times (-1) = -7$ means that the '7' changes sign, while the *absolute magnitude* is unchanged.[2] This rule of multiplication is consistent and holds for multiplication of any number.

Examples:

$(+12) \times (-1)$	$= -12$	$(+12) \times (-3)$	$= -36$
$(-2) \times (-3)$	$= +6$	$(-4) \times (+2)$	$= -8$

So far we have talked about multiplication, but what about *division*? Fortunately, division is very similar since it is the 'reverse' of multiplication and therefore the corresponding rules apply. If the two numbers that you are dividing have the *same sign* (positive or negative), the result is a positive number. On the other hand, if the two numbers that you are dividing each have *different signs*, the result will be negative. To conclude.

> Multiplication (or division) of *two positive*, or *two negative*, numbers gives a *positive* product. Multiplication (or division) of *one positive* and *one negative* number gives a *negative* product.

1.2 Brackets

It is often necessary to separate between numbers, and within equations, by brackets to avoid confusion and errors. The rule of thumb is to always start by analyzing/calculating within brackets. However, related to this is the more general rule regarding other steps in computing mathematical expressions. The convention is to start with brackets, then exponents, then division, then

[2] The absolute magnitude, or number, is written as $|7|$.

multiplication, then addition and subtraction.[3] As a very simple illustration of the importance of brackets, consider the following expression

$2 \times (2 - 3) = 2 \times (-1) = -2$

and compare it to an alternative (wrong) way

$2 \times 2 - 3 = 4 - 3 = 1$

which obviously renders a completely different result. Another illustrative example[4] is the difference between $4n^2$ and $(4n)^2$. Substitute $n = 2$ and you'll see that the first one results in 16 (4×4), while the second one results in 64 (8×8).

1.3 Fractions

A fraction[5] is when we divide one number by another such as e.g., $\frac{2}{3}$, $\frac{a}{b}$, $\frac{2+a}{b-3}$. The top part of a fraction is called *numerator*, while the bottom part is called *denominator*. It is of course possible to add-, subtract-, multiply- and divide fractions with each other although you need to be careful in some cases. The first thing to notice is that you do not change the value of a fraction if you manipulate both the numerator and denominator in the same way simultaneously. For example, if you divide or multiply both the numerator and denominator by, say, 4, the result remains the same, i.e.,

$$\frac{3}{6} = \frac{\frac{3}{4}}{\frac{6}{4}} = \frac{3}{6} = \frac{1}{2}, \qquad \frac{3}{6} = \frac{3 \times 4}{6 \times 4} = \frac{12}{24} = \frac{1}{2}$$

Adding or subtracting fractions is not a problem if you just remember that the denominators need to be the a common number (it is okay to transform the fractions so that this criterion is fulfilled). Specifically, you write

[3] Notice that with several, or nested, brackets, the rule is to start with the innermost bracket and then work your way out.

[4] Recall that it is rather common, and fully okay, to skip the multiplication sign if we are multiplying by letters. For example, $A \times B$ is written as AB, or $z \times 3$ is written $3z$.

[5] A fraction may also be called ratio or quotient.

the fractions with a common denominator and add (or subtract) their numerators. Turning to multiplication of fractions, this is easier than addition or subtraction. To multiply fractions you just multiply their corresponding numerators and denominators. Finally, division of fractions is very similar to multiplication although you need to turn the divisor upside down before multiplying (the inverse). Finally, to express a fraction in its simplest form you look for the highest common factor of the numerator and denominator. You then divide both the top and bottom of the fraction with this number.

Examples

$$\begin{aligned}
\frac{1}{3}+\frac{1}{10} &= \frac{1\times 10}{3\times 10}+\frac{1\times 3}{10\times 3}=\frac{(10+3)}{30}\\
\frac{1}{3}\times\frac{1}{10} &= \frac{1\times 1}{3\times 10}=\frac{1}{30}\\
\frac{\frac{1}{3}}{\frac{1}{10}} &= \frac{1}{3}\times\frac{10}{1}=\frac{10}{3}\\
\frac{36}{6} &= \frac{(36/6)}{(6/6)}=\frac{6}{1}=6
\end{aligned}$$

1.4 Basic Algebra

As soon as we use letters and other symbols to represent specific sets of numbers, values, etc. we have an algebraic expression. Algebra is simply calculation, or counting, with letters instead of numbers (although numbers may be part of an algebraic expression). In algebra, letters are typically used to reflect numbers which means that the standard mathematical rules apply also for letters. Algebraic expressions are often very convenient when you want to be more *general* in describing relations of facts or when you want to shorten the expression. For example, instead of saying that you pay ‘the salary’$\times 0.33$ in taxes you may define the salary as A, the tax payment as T

and t to be the tax rate. That is, $T = t \times A$.[6]

1.5 Integer (and fractional) Powers

Certain numbers (and algebraic expressions) are very convenient to write with *integer powers*. For example, 9 may be written as 3^2 (3×3) and 1000 may be written as 10^3 $(10 \times 10 \times 10)$. Specifically, a *product* $(a \times a)$ can be written as a^2 since you are allowed to add *exponents* for a product with identical *base* (in this case a) - we say that the base a is raised to two. The logic behind the 'two' in this case is that something raised to 'one' is 'itself' (i.e. $a \times a = a^1 \times a^1 = a^{1+1} = a^2$). In general terms, if we define a as the base and n as the exponent, the expression a^n is called the nth power of a. An application of integer powers is compound interest. Starting with 1000kr on you bank account, how much is it worth after, say, 5 years? Assume that you start with 100kr and that the interest rate is 3%. After one year you have $100 \times (1 + 0.03)$, after two years you have $100 \times (1 + 0.03) \times (1 + 0.03)$, etc. up to five years. However by using integer powers this can be written in a much more convenient way such that $100 \times (1 + 0.03)^5 = 100(1.03)^5 \approx 116$.

Exponents can be both positive and negative. Negative exponents follow the convention that $a^{-n} = \frac{1}{a^n}$. A typical application of the use of a negative exponent arises when we think about earning interests on a bank account. Given that I would have liked to have 100000kr on my bank account today, how much should I have deposited 10 years ago with an annual interest rate of 15%? By denoting the amount required as X, we realize that $X \times (1.15)^{10}$ must equal 100000kr.[7] Rearrangement gives that

$$X = \frac{100000}{1.15^{10}} = 100000 \times 1.15^{-10} \approx 24718kr$$

[6] In what follows we often simplify the notation by skipping the '$\times$' sign such that e.g. $a \times b = ab$.

[7] $X \times 1.15 \times 1.15 \times 1.15... = 100000$

In some cases we find powers with a fractional exponent. An example could be a production function with capital and labor as inputs[8], $Y = K^{2/5}L^{3/5}$. Another application of a fractional exponent is the *square root.* The square root can be written as $\sqrt{a}$, or, equivalently, as $a^{1/2}$.

There are of course more examples and applications of the use of integer powers and all of them are not mentioned here. However, below you find a summary of the most common rules and properties of powers.

Rule	Example
$a^0 = 1$	$234^0 = 1$
$a^{-n} = \frac{1}{a^n}$	$12^-2 = \frac{1}{12^2} = \frac{1}{144}$
$a^n \times a^m = a^{n+m}$	$2^2 \times 2^3 = 2^5$
$\frac{a^m}{a^n} = a^{m-n}$	$\frac{3^2}{3^3} = 3^{-1}$
$(a^n)^m = a^{nm}$	$(4^2)^3 = 4^6$
$(ab)^n = a^n b^n$	$(5 \times 2)^2 = 5^2 2^2 = 100$
$(\frac{a}{b})^n = \frac{a^n}{b^n}$	

1.6 Exercises

1. Evaluate.

   (a) $(-2)8(-1) + 12/2 + 3 - 4$

   (b) $\frac{5(-4)(-1)(-3)}{(-6)2}$

   (c) $-68 - (-62)$

2. Simplify.

   (a) $2x + 6y - x + 3y$

   (b) $8r^2 + 4s - 6rs - 3s - 3s^2 - 7rs$

   (c) $2e^2 + 5f - 2e^2 - 9f$

[8] Recall that an economist would say that the production function has a constant returns to scale technology when the exponents add up to one (as in this case).

3. Multiply out the brackets and simplify.

   (a) $x(2-x)$

   (b) $4(x-2)-3x-z+z^2$

   (c) $z-x+12-(x^2+10-4)$

   (d) $(x+2)(1-x)$

4. Reduce each of the fractions to its lowest terms.

   (a) $\frac{12}{9}$

   (b) $\frac{36}{48}$

   (c) $\frac{3x}{4x}$

   (d) $\frac{x^2}{x}$

   (e) $\frac{4x}{2x^3}$

   (f) $\frac{4c}{2a+4x}$

   (g) $\frac{z(z+3)}{z(1-z)(z+3)}$

5. Reduce to integers or fractions with lowest terms.

   (a) $\left(\frac{1}{3}\right)^4$

   (b) $\left(\frac{1}{4}\right)^{-2}$

   (c) $\left(-\frac{1}{3}\right)^4$

6. Simplify the following algebraic expressions.

   (a) $\left(\frac{2}{xy}\right)/\left(\frac{x}{y}\right)$

   (b) $\left(\frac{2x}{6y}\right)/\left(\frac{x}{3y}\right)$

   (c) $2x^2+x^3+4x-x^2+y^3$

Solutions 1. (a) 21 (b) 5 (c) -6

2. (a) x+9y (b) $8r^2 + s - 13rs - 3s^2$

3. (a) $2x - x^2$ (b) $x - 8 - z + z^2$ (c) $z - x + 6 - x^2$ (d) $-x - x^2 + 2$

4. (a) $\frac{4}{3}$ (b) $\frac{3}{4}$ (c) $\frac{3}{4}$ (d) x (e) $2x^{-2}$ (f) $\frac{2c}{a+2x}$ (g) $\frac{1}{1-z}$

5. (a) $\frac{1}{81}$ (b) 16 (c) $\frac{1}{81}$

6. (a) $2x^{-2}$ (b) 1 (c) $x(x + x^2 + 4) + y^3$

2 Equations

An equation is, loosely speaking, a relationship that can be solved mathematically. For example, the expression $2x + 3x = 5x$ defines an *identity*, while $5x = 10$ defines an equation. The difference is that, in the latter, you may solve for x and find that $x = 10/5 = 2$. Equations can of course take on a variety of forms and the general method is to collect the unknown variable (the x in the previous equation) on one side of the equation and then check what is left on the other side. A general type of equation in economics and econometrics is $y = kx + m$ where x and y are the variables, and k and m are so called *parameters*.[9] There is a specific terminology such as *endogenous variables*, *exogenous variables*, *structural form* and *reduced form*. To clarify, let us start from the well known GDP equation $Y = C + I + G$ where Y is GDP, C consumption, I investment and G government spending. In addition, assume that aggregate consumption is defined as $C = c_0 + cY$ where c_0 is autonomous consumption and c is the marginal propensity to consume of income (GDP), whereas I and G are treated as fixed. Given this information we are able to solve for Y by substituting the equation for consumption to the GDP equation,

$$Y = c_0 + cY + I + G$$

[9]Notice that the letters are arbitrarily chosen and of no importance as such.

Rearranging so that all terms containing Y are on the left hand side gives

$$Y - cY = c_0 + I + G \quad \Rightarrow \quad Y(1-c) = c_0 + I + G \quad \Rightarrow \quad Y = \frac{c_0}{1-c} + \frac{I+G}{1-c}$$

Thus, the *endogenous* variable Y is now expressed in terms of the *exogenous* variables I and G, and the parameters c and c_0. Substituting values of the constants, I, G, c, c_0, in this expression gives the value of GDP. Finally, in economics we say that the initial equations (defining Y and C) are the *structural* form of the model, while the final equation for Y is one part of the *reduced* form of the model of GDP.[10]

Example Solve the quadratic equation $4x^2 = b$.

Solution The first step is to divide both sides by 4 so that $x^2 = b/4$. The second step is to try to get rid of the 'squared', which can be done if we use fractional powers and raise both sides by 1/2. Given an example where $b = 16$, the x is solved for as follows

$$x^2 = 16/4 = 4 \quad \Rightarrow \quad (x^2)^{1/2} = 4^{1/2} \quad \Rightarrow \quad x = \sqrt{4} = \pm 2$$

The trick to use fractional powers is very convenient and works in a similar fashion for higher order equations like e.g. the cubic equation $x^3 = b$.

2.1 Exercises

1. Solve the following equations.

   (a) $2x + \frac{x-400}{3} = 100$

   (b) $\frac{5}{6} - \frac{x}{2} - \frac{x}{8} = 0$

   (c) $\frac{3x-1}{2} = \frac{2x-1}{3}$

[10] The other part of the reduced form would be if we expressed the other endogenous variable, C, in terms of the exogenous variables and the parameters.

(d) $\frac{x}{12} = \frac{x}{21} - \frac{x-1}{28}$

(e) $42 = 0.6(x + 30)$

(f) $4(1 - 0.3x) = 2.2$

Solutions 1. (a) x=100 (b) $x = \frac{4}{3}$ (c) $x = 1/5$ (d) $x = 0.5$ (e) $x = 40$ (f) $x = 1.5$

3 Functions of one variable

The mathematical term and concept *function* is widely used within economics. Examples are production functions, demand- and supply functions, cost- and profit functions, consumption functions etc. Basically, a function describes simple as well as complicated relationships between variables. In some cases we think of a relationship where one variable is a function of (depends on) another variable, whereas in other cases one variable can be a function of several variables. For a standard macroeconomic production function we often say that total production depends on the level of capital and labor, i.e. it is a function of capital and labor (two variables). If we know (or think we know) the exact relationship we can write what is called an *explicit* function[11] such as e.g. $Y = 0.5K + 4L^2$. If we don't know the exact relationship we usually write a *general* function like e.g. $Y = F(K, L)$.[12] In addition, we often use the words *dependent* and *independent* variables. In the example above, Y is the dependent variable while K and L are the independent variables - Y depends on K and L which are considered as 'independent'. In economics, we often use the word *exogenous* (fixed outside

[11] A function in which the dependent variable can be written explicitly in terms of the independent variable(s).

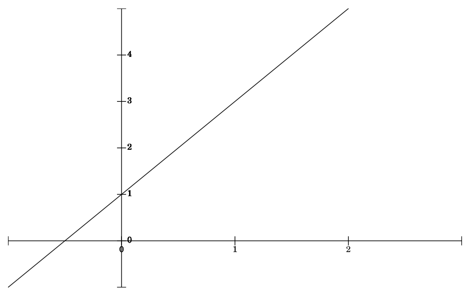
[12] The 'F' works to indicate 'a function of'. Although F and f are common notation, the letter or symbol as such does not matter and we are free to use whatever we want to denote a function.

the economic model) to denote the independent variables, whereas the word *endogenous* (determined within the economic model) is used for the dependent variable. Throughout the rest of this section, focus will be on functions with just one independent variable, typically written as $y = f(x)$.

To better understand the meaning of a specific functional form, we often illustrate functions graphically to interpret and describe the properties of the function. The dependence between two variables can be illustrated by a *curve* in two dimensions. Given the $y = f(x)$ form, the dimensions are x, y, where (by convention) y is the vertical axis and x is the horizontal axis. Starting from a pre-specified (explicit) function it is rather straightforward to make a plot by substituting for different values of the exogenous variable (x) to get corresponding values for the endogenous variable (y). Given the function $y = 2x + 1$ as an example, we are able to calculate the pairs of x's and y's as

x	0	1	2
$y = 2x + 1$	1	3	5

which then can be plotted in a graph like

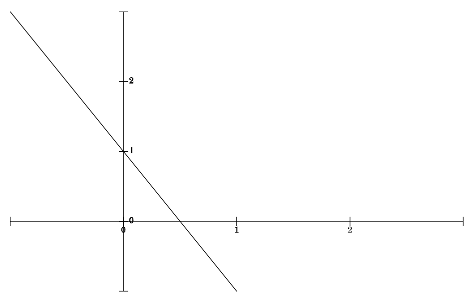


$y = 2x + 1$

Note that it would actually be enough to calculate two pairs of x and y (linear function). In what follows, you find examples of common functional forms and their corresponding properties, interpretation and graphical illustration.

3.1 Linear functions

Think about a job where you get payed on hourly basis. In that case, your total (gross) salary is linear in the hours of work since every hour increases the salary by the same amount. Total salary is your dependent variable, y, and hours of work is your independent variable, x. By defining the wage rate as k, total salary can be written as $y = kx$. The word 'linear' comes from the fact that the relationship between total salary and hours of work can, in this example, be illustrated by a straight 'line'. To be somewhat more general we often define a linear function as $y = kx + m$, where m is the *intercept*, or constant (equal to zero in the salary example). Changing to a macroeconomic example of total consumption where $C = c_0 + cY$, the intercept is c_0 and describes the consumption level when income (Y) equals zero. Translating this into the general form, the intercept tells us the value of the dependent variable when the independent variable equals zero. Next, the linear function $y = -2x + 1$ is illustrated as (recall that you could do this manually in the same way as for the graph above).



$y = -2x + 1$

It turns out rather obvious that this straight line has a very different slope compared to the previous graph. However, these functions only differ with respect to one thing, the $+2x$ and the $-2x$. The $+2$ and the -2 are said to be the *slope coefficients* for each respective function. For the general case, the

'k' is the corresponding slope coefficient. The slope coefficient measures the steepness and determines whether the line should point up- or downwards. A larger number means 'steeper' and a negative number means downwards, and vice versa. In general, the slope coefficient tells us how the dependent variable changes from a one unit increase in the independent variable.

Starting from a straight line, how to calculate the corresponding functional form? It turns out that two points on the straight line are enough to find the functional form. Think of a straight line passing through a point (x_1, y_1) with slope k. Given any other point on the line, (x, y), the slope will be defined by the change in one dimension (y), relative to the change in the other dimension (x), i.e.

$$k = \frac{y - y_1}{x - x_1}.$$

Multiplying both sides by $(x - x_1)$ and rearranging give us the equation for a straight line, $y - y_1 = k(x - x_1)$.

Example Assume that we know two points on the line, (x_1, y_1)=(1,1) and (x_2, y_2)=(3,2). Find the corresponding linear function.

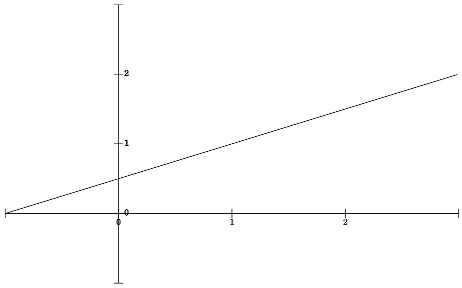
Solution As mentioned above, the slope coefficient defines how y changes when x changes, i.e.

$$k = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1}{2} = 0.5.$$

Given the equation for a straight line, the slope coefficient ($k = 0.5$) and one of the two points (does not matter which) we find that

$$y - 1 = 0.5(x - 1) \quad \Rightarrow \quad y = 0.5x + 0.5,$$

which is the explicit function defining the straight line.



$y = 0.5x + 0.5$

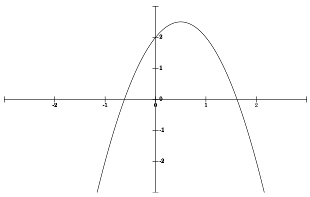
Although the linear form is fundamental and elementary in economics it is not necessarily relevant in all cases. Still, linear functions are often used as more or less suitable approximations in economics. The supply and demand curves are sometimes illustrated as linear in economic models although they are not necessarily so in real world. For example, it is likely that a supply function has a positive slope that becomes steeper when the quantity increases (almost vertical at some point), which means that a linear relationship is not perfect.

3.2 Quadratic functions

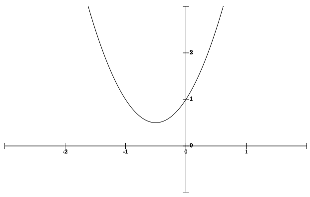
From introductory microeconomics we know that the average cost curve facing a firm is more or less U-shaped, and a linear function is therefore not a good choice. Another example is a profit function that typically has some maximum point (value) that the firm tries to 'find'. Given a relationship that either decrease to some minimum level, or increase to some maximum level, a *quadratic* function could be an appropriate choice. The quadratic function is defined as

$$f(x) = ax^2 + bx + c \qquad (a, b \text{ and } c \text{ are constants})$$

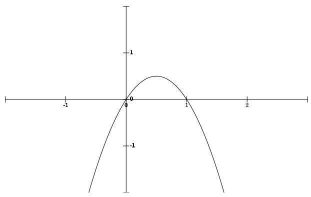
Notice that if $a = 0$ the quadratic function turns to a linear function. Therefore, to be a quadratic function we need that $a \neq 0$. The shape of the general quadratic function resembles a $\cup$ when $a > 0$ and a $\cap$ when $a < 0$. This also shows that the graph of a quadratic function is symmetric around its maximum or minimum. Below you find examples of quadratic functions.



$y = -2x^2 + 2x + 2$



$y = 2x^2 + 2x + 1$



$y = -2x^2 + 2x$

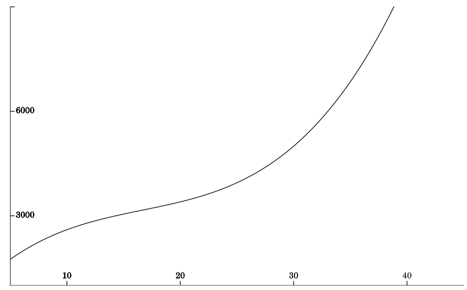
Although the general shape is straightforward, it is interesting to find that (i) the constant, c, gives us the intercept of the y-axis, and (ii) the quadratic function has two 'zeros' in two of the examples. The x-values that gives $y = 0$ are found by solving the equation for $y = 0$ (the 'quadratic formula' in Sydsæter 2012, p. 43). By this method it can be shown that the function will have two zero values when $b^2 - 4ac \geq 0$ (this is however not within the scope of this text).

3.3 Polynomials

A constant $y = a_0$ is a polynomial of degree zero; a linear function $y = a_1x + a_0$ is a polynomial of degree 1; a quadratic function $y = a_2x^2 + a_1x + a_0$ is a polynomial of degree 2 and a cubic function $y = a_3x^3 + a_2x^2 + a_1x + a_0$ is a polynomial of degree 3. The lesson from these statements is that the word *polynomial* has to do with the highest nonnegative integer power (exponent) of x in a function (besides its direct translation to 'multiterm'). In more general notations, a polynomial is written

$$f(x) = a_nx^n + a_{n-1}x^{n-1} + ... + a_1x + a_0 \qquad (a's \quad \text{are constants and} \quad a_n \neq 0).$$

Higher order polynomials are not that common within economics although the cubic function can occasionally represent cost functions. A firm often starts with a fixed cost. The increased production increases the cost, although at a decreasing rate when economics of scale kicks in. At some level of production, it is possible/likely that the cost increases at a faster rate again, because of e.g. technical bottlenecks and increased over time payments. This story of cost could be represented by a cubic function like,



$$y = 0.4x^3 - 20x^2 + 400x + 200$$

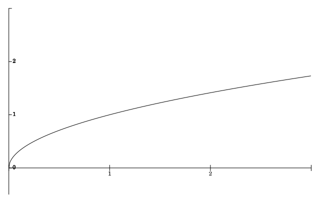
3.4 Power functions

A power function is somewhat simpler to graph and understand than the higher order polynomials. A power function is generally defined as

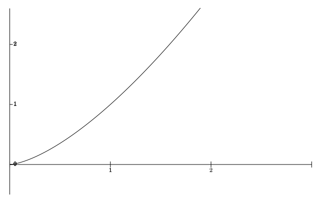
$$y = Ax^k$$

where k can be any rational (fractional) number, $x > 0$ and A is an arbitrary constant.[13] An application of a power function in economics is the well known Cobb-Douglas production function defined as $q = AK^aL^b$, where A, a and b are constants. Recall that the properties of the Cobb-Douglas production function largely depend on the values of the exponents. For example, if the exponents sum to one ($a+b=1$) the production is characterized by constant returns to scale technology (it is a good exercise to prove this statement). Below you find three graphs of power functions.

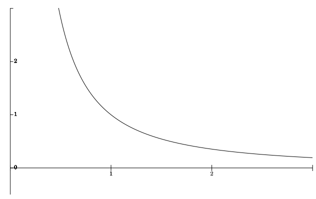
[13]For some exponents, such as e.g. $k = 1/2$, the function is not defined for negative values of x.



$y = x^{0.5}$



$y = x^{1.5}$



$y = x^{-1.5}$

Notice that all the graphs pass through the point $(x, y) = (1, 1)$ since $f(1) = 1^k = 1$, and that the shape depends on the value of k.

3.5 Exponential functions

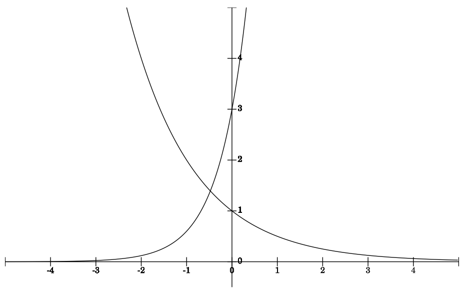
It is easy to be confused by the difference between power functions and exponential functions. However in theory, the difference is very well defined since the exponential function has the 'x' in the exponent, while the power function has the 'x' as its base.[14] Accordingly, the exponent varies in the exponential function. The exponential function is written

$$y = Aa^x$$

where a and A are positive constants. Exponential functions are often used to describe economic relationships over time such as e.g. economic growth, population growth and the money growth (hopefully) on our bank account. The value of the *base* determines whether the function is increasing, decreasing or constant. If (i) $a > 1$ the function is increasing, if (ii) $0 < a < 1$ the function is decreasing and if (iii) $a = 1$ the function is constant and equal to A. See below for two exponential curves (in the same figure).

Notice that the constant, A, determines the intercept while the base determines the slope. A large exponent in absolute numbers means a steep slope, a positive (negative) exponent means a positive (negative) slope.

[14] Notice the difference between $y = Aa^x$ and $y = Ax^a$.



$y = 3 \times 5^x, \quad y = 2^{-x}$

Example Today an Iphone costs about 5500kr. If the price of an Iphone follows the inflation rate of 3%, how much will it cost in 5 years, 10 years, 15 years?

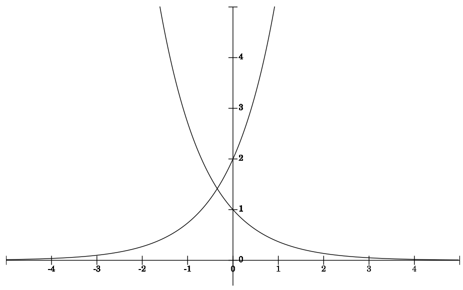
Solution To answer the question it is convenient to use the method of exponential functions. Use the general exponential function above and substitute such that y is the price of an Iphone, $A = 5500$, $a = (1 + 0.03)$ and x is the number of years.
$y = 5500 \times 1.03^0 = 5500\text{kr} \quad \text{(today)}$
$y = 5500 \times 1.03^5 \approx 6376\text{kr}$
$y = 5500 \times 1.03^{10} \approx 7391\text{kr}$
$y = 5500 \times 1.03^{15} \approx 8569\text{kr}$

3.5.1 Natural exponential functions

The exponential function is defined by the value of its base. In mathematics some bases are more used than others. One of the most widely used bases is the number 2.71828 which is called e in mathematics. The reason behind its popularity in mathematics is that calculation often simplifies (to be seen in what follows). The natural exponential function is written

$$y = Ae^x.$$

By looking at the graphs illustrating the national exponential functions below (same figure) you soon find the resemblance to other exponential functions, which is no surprise since e is just a pre-specified base.



$$y = 2e^x, \quad y = e^{-x}$$

Notice once again that the constant ('A') determines the intercept and that a negative exponent means a 'downward' sloping curve. The natural exponential function is typically motivated by calculus convenience.

3.6 Logarithmic functions

Exponential functions are often used to characterize processes over time, such as growth rates. Taking it the other way around, exponential functions can be solved to answer questions like: Starting with 1000kr, *when* do I have 100000kr on my account if the interest rate is 8%? The answer is that we need to solve for t (time) in the equation $1000 \times (1 + 0.08)^t = 100000$. The solution to this problem is much easier to find by the use of 'e' (see below).

In general notation, we would like to solve the equation $Aa^t = b$ (the exponential function above) where the base, a, is now equal to e ($= 2.71828$). Assuming that $A = 1$ and $a = e$ (i.e. $e^t = b$), for each number $b > 0$ there exists a number t such that $e^t = b$. The number t is the *natural logarithm* of

b, which is written $t = ln(b)$. In other words, we can write

$$e^{ln(b)} = b.$$

There are some basic rules that apply for logarithms.[15] These are summarized below.

$$\begin{aligned}
&\text{(i)} \quad ln(xy) = ln(x) + ln(y) \\
&\text{(ii)} \quad ln\frac{x}{y} = ln(x) - ln(y) \\
&\text{(iii)} \quad ln(x)^k = kln(x) \\
&\text{(iv)} \quad ln(1) = 0, \qquad ln(e) = 1
\end{aligned}$$

These definitions and rules are very important and should be memorized for future reference. Let us now return to our example and check whether we are able to use these rules to solve our problem. Taking the logarithms of both sides and then rearranging give the following

$$\begin{aligned}
1000 \times (1.08)^t &= 100000 \\
ln\big(1000 \times (1.08)^t\big) &= ln\big(100000\big) \\
ln1000 + ln(1.08)^t &= ln100000 \\
ln1000 + t \times ln(1.08) &= ln100000 \\
t \times ln(1.08) &= ln100000 - ln1000 \\
t = \frac{ln100000 - ln1000}{ln(1.08)} = \Big(\frac{ln(100000/1000)}{ln(1.08)}\Big) &\approx 60
\end{aligned}$$

That is, it will take close to 60 years for the 1000kr to become 100000kr if the interest rate is 8 percent.

3.7 Exercises

1. Determine the function of the straight line passing through (x, y)...

[15]Besides the natural logarithm, the logarithm with a base equal to the number 10 is the most common. The '10th' logarithm is often denoted just log or log_{10}.

(a) $(-4, 3)$ and $(6, 5)$

(b) $(1, -2)$ and $(-3, 6)$

(c) $(0, 2)$ and $(10, 2)$

2. Use the rules of logarithms to expand.

   (a) $ln(xy)$

   (b) $2 + lnx^3$

   (c) $lnxy^6$

   (d) $ln\frac{y}{z}$

   (e) $ln\frac{x^2}{y^3}$

   (f) $ln(xyz)$

3. Express each of the following as a single logarithm.

   (a) $lnx + lny$

   (b) $3lnx + lnx$

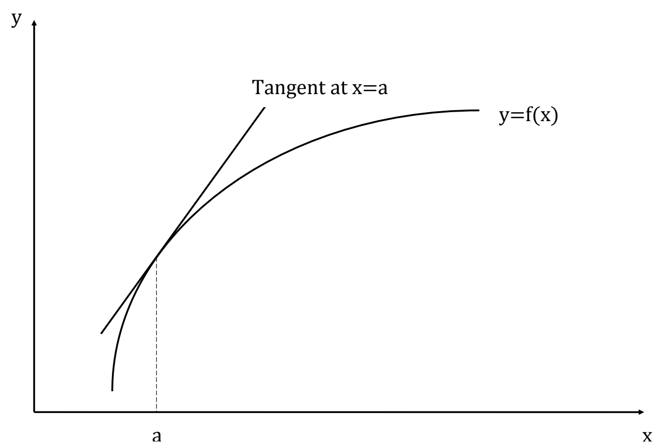
   (c) $2lnx - lny$

Solutions 2. (a) $lnx + lny$ (b) $2 + 3lnx$ (c) $lnx + 6lny$ (d) $lny - lnz$ (e) $2lnx + 3lny$ (f) $lnx + lny + lnz$
3. (a) $lnxy$ (b) lnx^4 (c) $ln\frac{x^2}{y}$

4 Differentiation

Given an economic relationship, or function, it is often of interest to study the effects on the dependent variable from a change in the independent variable(s). For example, how will profit in a firm change from a small change in the labor force? The answer to this type of question has to do with the slope of the curve illustrating the function of interest. A *derivative* is the mathematical term for this measure. The derivative describes/defines what happens to the function on the 'margin' (marginal change in the independent variable). The method to calculate the derivative is called *differentiation*.

Thinking about a linear function we have actuallt discussed how to calculate the derivative in a previous section. The derivative is equal to the slope, which is identical for all points on a linear function. However for a non-linear function, things get a bit more complicated since the slope differs depending on where we are on the 'curve'. The idea underlying the derivative concept is that, in every point at the curve, there exist some straight line touching the curve. The straight line just touching the curve is called the *tangent* to the curve in that specific point. Hence, if we were able to calculate the slope of the tangent we would also know how the curve changes in that exact point, i.e. the derivative of the curve (function). Below you find a graph to illustrate the idea.



It is worth noting that the slope in this example is positive and decreases (flatter) as x increases. In mathematics, the symbol $f'(a)$ (in words *'f-prime'*) represents the slope of a function f at the point where $x = a$. Since each value of x defines a value of the derivative (slope), the slope is in turn defined by a function (derived function) that is written

$$\frac{dy}{dx}$$

Note that this does not mean that you divide dy by dx, it is more like a symbol for the 'derivative of y with respect to x'.[16] However notice that it does not really matter if you write $\frac{dy}{dx}$ or $f'(x)$ since you can, for example, write $y = f(x) = x^2$.

In practice you could try to manually sketch tangents and calculate the corresponding slopes . However such procedure is hard to do with any precision - you have to draw the curve correct, find the exact tangent and measure the tangent's slope correctly. Fortunately there exists a very convenient mathematical method to find the derivative $f'(a)$. The method of finding the derived function (derivative) is called *differentiation*. The first thing to know is the simple formula to find $f'(a)$ when f is a power function - *the power rule.*[17]

$$\begin{aligned} \text{If} \quad & f(x) = x^n \Rightarrow \quad f'(x) = n \times x^{n-1} \\ & \text{or equivalently} \\ \text{If} \quad & y = x^n \Rightarrow \quad \frac{dy}{dx} = n \times x^{n-1} \end{aligned}$$

The power rule is extremely important and is essential for all types of differentiation. Variations in power functions are generally very straightforward to differentiate.

[16] The logic behind the notation is that the 'change' is historically denoted as the Greek letter Δ (delta), which corresponds to 'd' in English.

[17] The proof of this formula is not within the scope of this course, but can be found in most math textbooks.

Examples of the power rule

$$y = x^3 \qquad \rightarrow \qquad \frac{dy}{dx} = 3x^2$$
$$y = -x^3 \qquad \rightarrow \qquad \frac{dy}{dx} = -3x^2$$
$$y = x^{-2} \qquad \rightarrow \qquad \frac{dy}{dx} = -2x^{-3}$$
$$y = \sqrt{x} = x^{1/2} \qquad \rightarrow \qquad \frac{dy}{dx} = 0.5x^{-1/2}$$

4.1 Rules of differentiation

Although the power rule is fundamental, more sophisticated problems demand additional rules. Below you find brief presentation of differentiation rules that certainly are within the scope of this course. Notice that the rules not just work separately but can also be combined to find the derivative for more complex functions than in the examples.

The constant rule If you have a function that is multiplied by a constant, you differentiate the function and multiply by the constant. That is,

$$\text{if} \quad g(x) = Af(x) \quad \text{then} \quad g'(x) = Af'(x)$$

But what if you just have a constant, not multiplied by a function? Note that a constant can be rewritten to include x such that

$$y = A \quad \Rightarrow \quad y = Ax^0$$

which can then be differentiated in line with the constant rule.

$$\text{if} \quad y = Ax^0 \quad \text{then} \quad \frac{dy}{dx} = A \times 0 \times x^{-1} = 0$$

Hence, the derivative of a constant is zero.

The sum and difference rule If you have a sum (difference) of two functions, differentiate each function separately and add (subtract). That is,

$$\text{if} \quad g(x) = h(x) + f(x) \quad \text{then} \quad g'(x) = h'(x) + f'(x)$$
$$\text{if} \quad g(x) = h(x) - f(x) \quad \text{then} \quad g'(x) = h'(x) - f'(x)$$

The chain rule In many cases we are faced with functions that demand more sophisticated rules of differentiation. For example, how to differentiate $y = (2x + 3)^{10}$ with respect to x? The trick in many situations is to define 'new' functions within a function and then use the *chain rule*. To find some intuition, think about a case where y is a function of u, which, in turn, is a function of x. What happens to y when x changes? The logic answer is that x affects u, which affects y. In other words, there will be a chain reaction starting from the x. More formally we define the chain rule as

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Now, recall our previous example. By defining a function $u = (2x + 3)$, the function $y = (2x + 3)^{10}$ can be rewritten to read $y = u^{10}$. The first part of the chain rule is to differentiate the function y with respect to u, whereas the second part is to differentiate u with respect to x. Specifically,

$$\frac{dy}{du} = 10u^9, \qquad \frac{du}{dx} = 2$$
$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = 10u^9 \times 2 = 20u^9 = 20(2x + 3)^9$$

Note that the chain rule can be applied to longer 'chains' (more than two links).

The product rule In some cases functions are defined as a product of functions. For example, assume a function $F(x) = f(x)g(x)$, where $f(x) =$

x and $g(x) = x^2$. One approach is to multiply out the product $(x \times x^2)$ and differentiate, $F'(x) = 3x^2$. Notice however that this result is different compared to differentiation of the functions separately, i.e. $f'(x) \times g'(x) = 1 \times 2x$. So, which method is correct? The first gives the correct answer although there are an alternative and more general method to find the derivative - *the product rule*. The product rule is defined so that

$$F(X) = f(x)g(x) \quad \Rightarrow \quad F'(x) = f'(x)g(x) + f(x)g'(x) \tag{1}$$

If we apply the product rule to the example above it gives that $F'(x) = 1 \times x^2 + x \times 2x = 3x^2$, which corresponds to the above suggested.

The quotient rule How to handle a function which is a quotient of functions? The *quotient rule* defines that if

$$F(x) = \frac{f(x)}{g(x)} \quad \text{then} \quad F'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}. \tag{2}$$

Continuing with the previous example of $f(x) = x$ and $g(x) = x^2$, the differentiation of the quotient gives that $F'(x) = \frac{x^2 - 2x^2}{(x^2)^2} = -x^{-2}$. This result can be tested by using the product rule which, since $F(x) = \frac{x}{x^2} = x^{-1}$, suggests that $F'(x) = -x^{-2}$.

Exponential functions As mentioned above, exponential functions are popular for computational reasons and its derivative explains how. Without going into any details, the derivative of the exponential function is the function itself. Specifically,

$$\text{if} \quad f(x) = e^x \qquad \text{then} \quad f'(x) = e^x.$$

Note that if the exponent of e is a function itself (or even $-x$) we need to apply the chain rule. The simple rule to learn within this course is that if $f(x) = e^{ax}$, the derivative will be defined by $f'(x) = a \times e^{ax}$.

Logarithmic functions From a previous section we know about the relation between e and ln and it can be shown (not within the scope of this course) that

$$\text{if} \quad f(x) = lnx \qquad \text{then} \quad f'(x) = \frac{1}{x}.$$

4.2 Second order derivatives

The derivative itself produces a function which can be differentiated again. Thinking about slopes, the derivative of the slope will tell us how the slope itself changes for different values of x, i.e. we are talking the slope of the slope function. From a technical perspective this is straightforward and the same differentiation rules apply. The correct terminology is *second-order derivative*. The second-order derivative is written

$$f''(x) \qquad or \qquad \frac{d^2y}{dx^2}$$

and read 'dee two y, dee x squared'. Given that, for example, $f(x) = 3x^3 + 12x^2 - x$, the first-order derivative becomes $f'(x) = 9x^2 + 24x - 1$ while the second-order derivative becomes $f''(x) = 18x + 24$. The second order derivative can be used to determine whether a function is *convex* ($\cap$) or *concave* ($\cup$). For a convex function the slope starts out positive in order to decrease, become zero, and then turn negative. In other words, a convex (concave) function must correspond to a negative (positive) second order derivative - the slope of the slope function is negative (positive). Thus, the following holds

$$\text{The function } f \text{ is convex if} \quad f''(x) \geq 0$$
$$\text{The function } f \text{ is concave if} \quad f''(x) \leq 0$$

4.3 Exercises

1. Differentiate with respect to x.

   (a) $y = 3x^5 + 12x - 3x^2$

   (b) $y = (4x - 1)^2$

   (c) $y = (2x + 7)^6$

   (d) $y = (3x^2 + 2)^{126}$

   (e) $y = x(2x - 2)$

   (f) $y = (x - 1)(x + 3)^2$

   (g) $y = \frac{2x}{4}$

   (h) $y = \frac{x^2}{x^3}$

   (i) $y = \frac{x}{(x+2)}$

   (j) $y = \frac{x+3}{2x-2}$

   (k) $y = e^x - x^2$

   (l) $y = e^{6x}$

   (m) $y = lnx + 4$

   (n) $y = xe^x$

2. The price elasticity of supply is given by

$$E = \frac{\text{percentage change in supply}}{\text{percentage change in price}}.$$

   Given the supply function $Q = 124 - P^2$, find the price elasticity of supply at the point $P = 10$.

Solutions 1. (a) $15x^4+12-6x$ (b) $32x-8$ (c) $12(2x+7)^5$ (d) $756(3x^2+2)^125$ (e) $4x-2$ (f) $3x^2+10x+3$ (g) 0.5 (h) $-x^{-2}$ (i) $-2(x+2)^2$ (j) $-8(2x-2)^{-2}$ (k) e^x-2x (l) $6e^{6x}$ (m) $1/x$ (n) $e^x(1+x)$
2. $E=-8\frac{1}{3}$

5 Maximum, minimum and optimization with one variable

Finding the largest or smallest value is often of interest in economic contexts. Examples are the quantity that gives the largest profit or smallest cost, the tax rate that is most cost efficient, or the optimal number of employees in production. In the one variable case you can imagine the two-dimensional graph where you try to find the lowest or highest value for the function of interest. For a strictly ∪-shaped curve it is obvious that there will be just one *minimum* point, whereas in other cases there may exist several minimum points (e.g think about the cubic function). In analogy, we may also think of cases where we are trying to find *maximum* values.[18] In line with this reasoning we are thus searching for a value of the independent variable causing the function to have a zero rate of change. A zero rate of change coincides with a zero slope of the function, i.e. when the derivative equals zero. This conclusion actually leads us to the so-called *first-order condition* for an extreme point.

> Given that we are considering a function $f(x)$ that is differentiable at relevant intervals, for the value $x=c$ to be a maximum or minimum point of f, a *necessary condition* is that it is a *stationary* point, i.e. that the first-order derivative equals zero ($f'(c)=0$).

[18] Generally maximum or minimum is called extreme points.

The first-order condition is necessary but not *sufficient* to determine whether a maximum or a minimum. For example, irrespectively of a $\cap$- or $\cup$-shaped quadratic function it still has a point characterized by a zero slope. Considering a cubic function it becomes even more complex since it has several extreme points. Accordingly, extreme points may be *local* extreme points and we cannot determine whether they are maximum or minimum points from the first-order derivative. For some functions, such as e.g. the third-order polynomial (cubic function), there may also exist what is called *inflection* points where the slope is zero but it is neither a maximum nor a minimum. Specifically, an inflection point is where the graph rises on one side and falls on the other.[19] In most economic applications, inflection points are typically not of interest.

For most applications in economics, a local extreme point coincide with a global extreme point and we will therefore focus on such cases. Following the same reasoning as for the definition of convex and concave functions in a previous section, we know that the second-order derivative gives more information on the shape of a function. This leads us to the second-order derivative test for extremum.

> If the first derivative of a function $f(x)$ at $x = c$ is $f'(c) = 0$, then the value of the function at c will be
>
> - a relative *maximum* if the second-order derivative value at c is $f''(c) < 0$
> - a relative *minimum* if the second-order derivative value at c is $f''(c) > 0$

[19] There are additional definitions related to an inflection point although not relevant for now.

This test is very convenient and also sufficient in most economic applications although it has one important drawback. In some cases the second-order derivative value becomes zero, meaning that the test is inconclusive. If this is the case we need to revert to other tests that are not within the scope of this text. Leaving this aside, the second order test, often called the *second-order condition*, is sufficient to find maximum or minimum values.

Finding maximum or minimum points is often referred to as *optimization* in economics. To increase the understanding of optimization consider the following economic application.

Example Assume a demand function $Q = 40 - 2P$. Find the level of output, Q, that maximizes total revenue.

Solution Total revenue is simply the price times the quantity, $TR = PQ$, and we are looking for the value of Q that maximizes TR. First, we need to express TR in terms of Q. Specifically, the demand function can be rewritten as $P = 20 - 0.5Q$. Hence, total revenue is $TR = (20 - 0.5Q)Q = 20Q - 0.5Q^2$. We know that a stationary point is characterized by $\frac{d(TR)}{dQ} = 0$, meaning that $20 - Q = 0$. The quantity at the stationary point is therefore $Q = 20$. To determine whether this is a maximum or a minimum, we differentiate a second time to get $\frac{d^2(TR)}{dQ^2} = -1$. A negative second-order derivative indicates that we have found a maximum quantity at $Q = 20$ and the revenue function is convex.

6 Functions of more than one variable

Thinking about real world economic behavior, it is obvious that most relationships involve more than two variables. For example, the demand for

Volvo cars not only depends on its own price but also on the price of other brands, income, fuel economy, etc. As for macro level economics, we know that production depends on many inputs, technology etc. Accordingly, it is relevant to extend the concept of functions to include several (more than one) independent variables. Although it is possible to extend results and notation in what follows to include several variables, we will simplify notation by first and foremost discussing cases with two independent variables. Specifically, a function f of two variables x and y is a rule that defines a specific outgoing number (often denoted z). A function of two variables can be written

$$f(x,y) = 3x + x^2y$$

$$or$$

$$z = 3x + x^2y$$

The letters as such are of no importance and you might define the variables as e.g. $x_1, x_2, ..., x_n$. The terminology is analogous to the one-variable case in the sense that we still define independent- and dependent variables. Regarding the graphical illustration of a multi-variable function it demands more than two dimensions which makes it difficult to do by hand. The two-independent-variable case would demand three dimensions and could be thought of as a surface, like a mountain range.

In spite of the drawback of not being able to graphically describe multi-variable functions, we can still differentiate functions to better understand the characteristics. Given a function of two variables, $z = f(x,y)$, we can determine *two* first-order derivatives - one with respect to x, and one with respect to y. These are the *partial* derivatives - calculated with the other variable held constant. For the example above, the partial derivatives are defined as

$$\begin{array}{ccccc} \dfrac{\partial z}{\partial x} & \text{or} & \dfrac{\partial f}{\partial x} & \text{or} & f_x \\ \dfrac{\partial z}{\partial y} & \text{or} & \dfrac{\partial f}{\partial y} & \text{or} & f_y \end{array}$$

Notice that we are using 'curly' d's to represent a partial derivative. As in the one-variable case, the derivative is typically a function itself and we are able to calculate the second-order partial derivatives.

$$\begin{array}{ccccc} \dfrac{\partial^2 z}{\partial x^2} & \text{or} & \dfrac{\partial^2 f}{\partial x^2} & \text{or} & f_{xx} \\ \dfrac{\partial^2 z}{\partial y^2} & \text{or} & \dfrac{\partial^2 f}{\partial y^2} & \text{or} & f_{yy} \\ \dfrac{\partial^2 z}{\partial y \partial x} & \text{or} & \dfrac{\partial^2 f}{\partial y \partial x} & \text{or} & f_{yx} \\ \dfrac{\partial^2 z}{\partial x \partial y} & \text{or} & \dfrac{\partial^2 f}{\partial x \partial y} & \text{or} & f_{xy} \end{array}$$

The last two second-order partial derivatives are often called *cross-derivatives* and for all economic applications it holds that

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x},$$

meaning that the order of the partial differentiation is of no importance. Differentiating with respect to x then y gives the same result (expression) as differentiating with respect to y then x.

7 Optimization with more than one variable

Not surprisingly, the methods for finding maximum or minimum points of a function of two variables are very similar to those previously used in cases of one variable. There is however one fundamental difference; several-variable problems need to be divided into *unconstrained* and *constrained* optimization. The intuition is as follows. Think about a utility function of two goods,

$U(x_1, x_2)$. The optimization in this case is of course to try to maximize utility, which is ever increasing in both goods. The solution to this problem is however not finite since we can pick ever-increasing values of each of the two goods and the utility therefore can be increased without any bound. In practice, this does not occur since we are typically faced by some limitation in the amount of money to spend. By considering a restriction in our budget we are considering a constrained optimization problem. In other words, as soon as we impose some sort of restriction to our optimization problem, we are thinking of a constrained optimization.

7.1 Unconstrained optimization

There are some relevant cases in economics that are interpreted as unconstrained problems. One example is profit maximization that typically gives a finite solution without having any constraints. However notice that the profit function has an implicit constraint since it it is typically defined as the difference between revenue and cost, $\pi = TR - TC$. Normally there is a conflict between making revenue large and cost small.

For an unconstrained optimization problem, the steps of maximization (minimization) are almost analogous to the one-variable case. Given a function of two variables, $z = f(x, y)$, the stationary points are found by *simultaneously* solving the first-order partial derivatives functions[20]

$$\frac{\partial z}{\partial x} = 0$$
$$\frac{\partial z}{\partial y} = 0.$$

That is, write down the expressions for the first-order partial derivatives, equate to zero (zero slope), combine these two functions and solve for x

[20]Think of it as finding the top of a mountain in three dimensions.

and y. When it comes to deciding whether we have found a maximum or a minimum point, the two-variable case demands an additional condition.

If, after finding the stationary points (a, b),

- $f_{xx} > 0, f_{yy} > 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b) we have found a *minimum* at (a, b).
- $f_{xx} < 0, f_{yy} < 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b) we have found a *maximum* at (a, b).
- $f_{xx}f_{yy} - f_{xy}^2 < 0$ at (a, b), then (a, b) is not an extreme point (we have found a *saddle-point*).

Example Assume a market characterized by competitive producers, two goods with prices $P_1 = 1000$ and $P_2 = 800$. Total cost is defined by $TC = 2Q_1^2 + 2Q_1Q_2 + Q_2^2$ (Q is output). Find the profit maximum and the corresponding levels of Q_1 and Q_2.

Solution Competitive market implies that prices are fixed (do not depend on firm level output). The revenue, cost and profit are defined as

$$TR_1 = 1000Q_1 \quad \text{and} \quad TR_2 = 800Q_2$$
$$TR_{tot} = TR_1 + TR_2 = 1000Q_1 + 800Q_2$$
$$TC = 2Q_1^2 + 2Q_1Q_2 + Q_2^2$$
$$\pi = TR - TC = 1000Q_1 + 800Q_2 - 2Q_1^2 - 2Q_1Q_2 - Q_2^2$$

Differentiate the profit function with respect to the two outputs, equalize to zero and solve simultaneously to find the stationary point.

$$\frac{\partial \pi}{\partial Q_1} = 1000 - 4Q_1 - 2Q_2 = 0$$
$$\frac{\partial \pi}{\partial Q_2} = 800 - 2Q_1 - 2Q_2 = 0$$

By solving these two equations simultaneously we find $Q_1 = 100$ and $Q_2 = 300$ to be the stationary points. The next step is to check whether this solution is a maximum or minimum. The second-order conditions are

$$\frac{\partial^2\pi}{\partial Q_1^2} = -4 < 0, \quad \frac{\partial^2\pi}{\partial Q_2^2} = -2 < 0$$
$$\Big(\frac{\partial^2\pi}{\partial Q_1^2}\Big)\Big(\frac{\partial^2\pi}{\partial Q_2^2}\Big) - \Big(\frac{\partial^2\pi}{\partial Q_1 Q_2}\Big)^2 = 4 > 0$$

and imply that we have found a maximum. Finally, substitute the values of Q_1 and Q_2 to find the value of maximum profit to be $\pi = 170000$.

7.2 Exercises

1. The demand function of a good is $Q = 30 - P$ and the total cost function is $TC = 0.5Q^2 + 6Q + 7$.

   (a) What is the level of output maximizing total revenue?

   (b) What is the level of output maximizing profit. In addition, calculate MR and MC at the point of profit maximum.

2. Find the stationary points of the following functions

   (a) $f(x, y) = x^2 + y^2 - 3x - 3y$

   (b) $f(x, y) = x^2 + 3xy - 3x^2 - 3y^2 + 10$

   (c) $f(x) = 2x^2 - 8x + 5$

   (d) $f(x) = 2x^2 + 3x^2 - 24x + 4$

3. A firm produces two goods and the profit function is given by $\pi = 24Q_1 - Q_1^2 - Q_1Q_2 - 2Q_2^2 + 33Q_2 - 43$. Find the output levels that maximizes profit and confirm by the second order condition.

Solutions 1. (a) $Q = 15$ (b) $Q = 8$, $MR = MC = 14$

2. (a) Minimum, x=1.5 and y=1.5, f=-4.5 (b) Maximum, x=0 and y=0, f=10 (c) Minimum, x=2, f=-3 (d) Minimum, x=2.4

3. $Q_1 = 9$ and $Q_2 = 6$, second-order conditions are satisfied

7.3 Constrained optimization

In the previous section we discussed the method of solving optimization problems given two independent variables that were allowed to take any values (no constraint). In most real world examples we are however faced with some sort of constraint limiting our choice of the values of the variables. That is, the more realistic scenario is that we want to optimize some function $u(x, y)$, typically called the *objective function*, subject to some constraint $\phi(x, y) = m$. This is what we call a constrained optimization problem.

An example of the constrained optimization is when the function u is the utility and the function ϕ is the budget constraint, possibly defined as $px + y = m$ (x and y are goods, m is income and p the price of good x). By looking at the budget constraint we find that this problem can actually be defined as an unconstrained problem if we rewrite the budget constraint to $y = m - px$ and then substitute this into the utility function u. This results in an unconstrained optimization of the function $v(x) = u(x, m - px)$ with respect to the single variable x. The method of transforming constrained optimization problems to unconstrained problems is practical in some cases, but difficult or even impossible in other cases[21].

Example Find the maximum value of the objective function $z = -2x^2 + y^2$ subject to $y = 2x - 1$.

[21] Transforming to a constrained optimization by substitution is called the *substitution* method

Solution Substitute $y = 2x - 1$ into the objective function and simplify. This gives that $z = 2x^2 - 4x + 1$, implying that z is now a function of just one variable and we can use the method described above (unconstrained optimization). The stationary point is found where

$$\frac{dz}{dx} = 4x - 4 = 0 \quad \Rightarrow \quad x = 1$$

The second-order derivative is

$$\frac{d^2z}{dx^2} = 4 \quad \Rightarrow \quad \text{minimum}$$

Finally, the value of the objective function at this point is $z = -1$.

Fortunately, in cases where the substitution method becomes to complicated (or impossible) economists often use the *Lagrange multiplier method.* The Lagrange multiplier method is actually often used for very simple constrained problems since the so-called Lagrange multiplier turns out to have an important economic interpretation.

To illustrate, recall the objective function $u(x, y)$ and the constraint $\phi(x, y) = m$ and solve with the Lagrange method instead of the substitution method. The problem can be defined as

$$max(min) \quad u(x, y) \quad \text{subject to} \quad \phi(x, y) = m$$

The first step is to introduce a 'scalar' called the Lagrange multiplier and (often) denoted by the Greek letter lambda, λ. The Lagrange multiplier is associated with the constraint and the so-called *Lagrangian* will be defined as

$$\mathcal{L}(x, y) = u(x, y) - \lambda(\phi(x, y) - m)$$

Notice that the expression $\phi(x, y) - m$ must be equal to zero if the constraint is fulfilled, meaning that $\mathcal{L}(x, y) = u(x, y)$ for the values (x, y) that satisfy the constraint. Given the Lagrangian, the next step is to calculate the first-order partial derivatives for the two unknowns, x, y, and equate these to zero (analogous to previous methods of optimization) while the constraint is satisfied.[22] To recapitulate, the Lagrange method works as follows.

To find the solution for an optimization problem defined as

$$max(min) \quad u(x, y) \quad \text{subject to} \quad \phi(x, y) = m$$

(i) Define the Lagrangian as $\mathcal{L}(x, y) = u(x, y) - \lambda(\phi(x, y) - m)$ where λ is a constant.

(ii) Differentiate $\mathcal{L}$ w.r.t x and y and equate the obtained derivatives to zero. You now have three equations:

$$\begin{aligned} \mathcal{L}_x = u_x - \lambda\phi_x &= 0 \\ \mathcal{L}_y = u_y - \lambda\phi_y &= 0 \\ \phi(x, y) &= m \end{aligned}$$

(iii) Solve the *three* equations simultaneously for x,y and λ.

The three equations in (ii) are the so-called first-order conditions.[23]

Example Think of a student having a utility function $U(x, y) = x^2 - 3xy + 12x$ where x and y are two goods. At the same time, the student has to comply to a restricted budget $2x + 3y = 6$. Maximize the ulility.

[22] It is possible, and some prefer, to think of the λ as an additional decision variable and thereby equate the first-order partial derivative w.r.t λ to zero. This way of thinking will give the same solution in this case but may cause problems in other cases that are outside the scope of this text.

[23] Throughout we will assume that there is only a single optimum and that it is obvious from the context whether it is a minimum or maximum. The corresponding second-order conditions are quite complicated and not within the scope of this text.

Solution Define the Lagrangian to be $\mathcal{L} = x^2 - 3xy + 12x - \lambda(2x + 3y - 6)$. Differentiate w.r.t. x, y and don't forget the constraint.

$$\mathcal{L}_x = 2x - 3y + 12 - 2\lambda = 0 \tag{3}$$

$$\mathcal{L}_y = -3x - 3\lambda = 0 \tag{4}$$

$$2x + 3y - 6 = 0 \tag{5}$$

These three equations must be solved simultaneously (all three are true in optimum). There are more than one way to solve these equations and I have chosen to illustrate one. First, equation (4) gives that $\lambda = -x$ which, substituted to equation (3), gives that $4x - 3y + 12 = 0$. If you then add equation (3) to equation (4) you get that $6x + 6 = 0 \quad \Rightarrow \quad x = -1$. Substitute $x = -1$ to equation (5) and you find that $y = 8/3$. Finally, the value of utility in optimum is found by using that $x = -1$ and that $y = 8/3$.

Interpretation of λ By looking at the Lagrangian once again and thinking of m as a constant that is possible to change, calculating the partial derivative w.r.t. m, $\partial\mathcal{L}/\partial m = \lambda$. The interpretation must be that the value of λ tells us how the optimal value of the objective function changes with respect to changes in the constraint constant m. In economic applications the constant m usually denotes the available stock of some resource. The economic intuition for λ is that it measures the change in e.g. utility or profit that can be obtained from a small increase in the resource. Economists usually call λ the *shadow price* of the resource in mind.

7.4 Exercises

1. Consider the function $W = 2x^2 - 3xh + 2h + 10$ and the constraint that $h = x$.

(a) Maximize the objective function with the method of substitution.

(b) Maximize the objective function with the Lagrangian method.

2. Consider the function $U = 800z - z^2 + 1000y - 2y^2$ and the constraint that $z + y = 1000$

   (a) Maximize the objective function with the method of substitution.

   (b) Maximize the objective function with the Lagrangian method.

Solutions 1. (a) x=1, h=1 and W=11 (b) x=1, h=1 and W=11
2. (a) z=633.33, y=366.67 (b) z=633.33, y=366.67

8 Matrices

As an economist you sooner or later run into different forms of tables and systems of equations. The concept of tables was perhaps familiar to you before reading this text, while systems of equations were introduced when setting the first-order partial derivatives equal to zero in an optimization problem. For both these examples it often turn out to be convenient to use *matrices*. Matrices can be used to present data from tables in a more compact form (econometric models/specifications are typically expressed in matrix form), and systems of equations can be solved by methods related to matrices (matrix algebra). The focus in this section is to introduce matrix notation and elementary matrix algebra.

8.1 Terminology

Think of a supermarket selling milk, cheese, butter and cornflakes. These goods are sold to six customers (c1-c6) and the amount sold each weak can be summarized in the following table.

	Milk	Cheese	Butter	Cornfl.
c1	2	3	5	1
c2	1	3	0	1
c3	4	0	1	1
c4	3	1	3	2
c5	0	6	6	0
c6	0	5	4	1

Instead of writing this in table format, the matrix **A** below captures the same information.

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & 5 & 1 \\ 1 & 3 & 0 & 1 \\ 4 & 0 & 1 & 1 \\ 3 & 1 & 3 & 2 \\ 0 & 6 & 6 & 0 \\ 0 & 5 & 4 & 1 \end{bmatrix}$$

Generally, a matrix is an arrangement of numbers, variables or symbols into rows and columns, surrounded by a pair of brackets.[24] The individual numbers are referred to as *elements* and a matrix is made up of rows and columns. In this case, the matrix $\mathbf{A}$ has six rows and four columns. Matrix $\mathbf{A}$ is therefore of *order* 6×4. Specifically, a matrix of order $m \times n$ has m rows and n columns. As perhaps noticed, the convention is to denote matrices by capital letters in bold type and their elements by the corresponding lowercase letter. In addition, subscript notation is used to clarify the exact position within a matrix. That is, for the matrix $\mathbf{A}$ the a_{ij} stands for the element that can be found in row i and column j. In the matrix $\mathbf{A}$, $a_{23} = 0$ (row 2 and column 3). More generally, a 2×4 matrix $\mathbf{D}$ would be written

$$\mathbf{D} = \begin{bmatrix} d_{11} & d_{12} & d_{13} & d_{14} \\ d_{21} & d_{22} & d_{23} & d_{24} \end{bmatrix}$$

A matrix consisting of a single row, or column, is referred to as a *vector.* A vector is usually defined by bold lower case letters and a 'prime' if it refers to a row vector. For example,

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ \cdot \\ \cdot \end{bmatrix}$$

and

$$\mathbf{c}' = \begin{bmatrix} c_1 & c_2 & \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

[24]The 'type' of brackets (square or parenthesis) is not of importance.

There are some additional terms and special matrices that are good to know about before continuing: a *scalar*, the *identity* matrix and the *transpose* matrix. A *scalar* is very simple and refers to a single number, or constant, and not to any matrix in the conventional way. As will be shown below, it is a rather simple procedure to multiply a matrix by a scalar.

The *identity* matrix is denoted **I** and consists of the values 1 and 0 such that the diagonal running from the top-left to the bottom right contains only the 1's, while all other elements are 0's. For example,

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The *transpose* matrix is created by switching the rows and columns: row one becomes column one, row two column two etc. The transpose is denoted by a 'prime'. For example, the transpose of

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 3 & 0 \\ 4 & 0 & 1 \end{bmatrix}$$

is

$$\mathbf{A}' = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 3 & 0 \\ 5 & 0 & 1 \end{bmatrix}$$

8.2 Matrix algebra

As indicated above, matrices are not just about presenting data, equations, etc. in a compact form. Instead, the use of matrices are potentially a very convenient method to solve systems of equations and perform rather complex algebra. In general, the general rules for algebra do *not* apply for matrices and there are some basics that need to be learned: matrix addition, subtraction, multiplication and scalar multiplication.

Two matrices are possible, and rather straight forward, to add or subtract given that they are characterized by the same dimensions (rows and columns). Assume that you have two matrices, **A** and **B**, representing weekly sales.

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 3 & 0 \\ 4 & 0 & 1 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 5 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

By adding matrix **A** and **B** we find the sales over two weeks in time, **C**. The procedure for adding matrices of the same dimensions is to add each and every corresponding element in the respective matrix. That is,

$$\begin{aligned} \mathbf{C} = \mathbf{A} + \mathbf{B} &= \begin{bmatrix} 2 & 3 & 5 \\ 1 & 3 & 0 \\ 4 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 3 & 2 & 2 \\ 2 & 5 & 0 \\ 1 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2+3 & 3+2 & 5+2 \\ 1+2 & 3+5 & 0+0 \\ 4+1 & 0+2 & 1+1 \end{bmatrix} = \begin{bmatrix} 5 & 5 & 7 \\ 3 & 8 & 0 \\ 5 & 2 & 2 \end{bmatrix} \end{aligned}$$

As for matrix subtraction, it is analogous to the addition of matrises.

Multiplication of matrices is somewhat more complicated and does not really follow from standard rules. The simplest form of of multiplication is by a scalar, which is analogous to the rules for addition and subtraction. Specifically, given a scalar of e.g. 1.1 (an increase of weekly sales by 10 percent) would mean a multiplication of each element by 1.1. For example, multiplying matrix **C** by 1.1 would result in the following

$$\mathbf{D} = 1.1 \times \mathbf{C} = \begin{bmatrix} 1.1 \times 5 & 1.1 \times 5 & 1.1 \times 7 \\ 1.1 \times 3 & 1.1 \times 8 & 1.1 \times 0 \\ 1.1 \times 5 & 1.1 \times 2 & 1.1 \times 2 \end{bmatrix} = \begin{bmatrix} 5.5 & 5.5 & 7.7 \\ 3.3 & 8.8 & 0 \\ 5.5 & 2.2 & 2.2 \end{bmatrix}$$

The multiplication of two matrices requires a different approach. Let us start with vector multiplication and assume that we would like to multiply matrix **C** (e.g. sales) by the column vector **b** (e.g. the price of each product),

$$\mathbf{b} = \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix}$$

The procedure is then to multiply each row in the **C** matrix by the column vector **b** such that

$$\mathbf{C} \times \mathbf{b} = \begin{bmatrix} 5 & 5 & 7 \\ 3 & 8 & 0 \\ 5 & 2 & 2 \end{bmatrix} \times \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix} =$$

$$= \begin{bmatrix} 5 \times 2 + 5 \times 4 + 7 \times 3 & 3 \times 2 + 8 \times 4 + 0 \times 3 & 5 \times 2 + 2 \times 4 + 2 \times 3 \end{bmatrix} =$$

$$= \begin{bmatrix} 41 & 38 & 24 \end{bmatrix}$$

In this example, the resulting matrix corresponds to the total revenue from each consumer.

The multiplication of two matrices follows the same procedure as for the vector multiplication although it becomes slightly messier as the dimensions increase. To illustrate, let us define two 'general' matrices, **A** and **B**, as

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

In this case, the trick is to treat matrix **A** as having two separate rows (vectors) and then multiplying each row in **A** by each and every column in **B**. The result will then be

$$\mathbf{A} \times \mathbf{B} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} & a_{11}b_{13} + a_{12}b_{23} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} & a_{21}b_{13} + a_{22}b_{23} \end{bmatrix}$$

This method is applicable for any two matrices although the 'messiness' surely increases with the size of the matrices.

There are however caveats related to matrix multiplication. First, thinking about the multiplication procedure in the example above it is true that the first matrix must have the same number of columns as the second matrix has rows. If this is the case the matrices are said to be *compatible*. If the compatibility constraint is not met, it is not possible to carry out the multiplication. Second, the order of multiplication is crucial, meaning that (in general)

$$\mathbf{A} \times \mathbf{B} \neq \mathbf{B} \times \mathbf{A}$$

Still, the 'sequence' of multiplication is not of importance, i.e.

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = \mathbf{A} \times (\mathbf{B} \times \mathbf{C})$$

Third, the size of the matrix resulting from the multiplication is determined by the number of rows in the first matrix, and the number of columns in the second matrix. In the example above a 2×2 matrix is multiplied by a 2×3 matrix, giving a 2×3 matrix.

Before closing the section on matrices, we are now ready to describe systems of equations, or economic type-models, in matrix form. Let us, for a moment, think about a standard market equilibrium model characterized by a demand function, a supply function and an equilibrium condition,

$$\begin{aligned} Q_d &= 3 + 2P \\ Q_s &= 1 + 4P \\ Q_d &= Q_s \end{aligned}$$

By collecting the constants on the right hand side, the equations can be rewritten as

$$\begin{aligned} Q_d - 2P &= 3 \\ Q_s - 4P &= 1 \\ Q_d - Q_s &= 0 \end{aligned}$$

Given this format it is possible to write these equations in matrix form (the trick is to use that e.g., $Q_s \times 0 = 0$)

$$\begin{bmatrix} 1 & -2 & 0 \\ 0 & -4 & 1 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} Q_d \\ P \\ Q_s \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$$

These matrices can be written in short as

$$\mathbf{Ax=b}$$

where $\mathbf{A}$ is the coefficient matrix, $\mathbf{x}$ the vector of variables and $\mathbf{b}$ the vector of constants.

The methods learned so far will prove to be very useful for future classes in economics. For example, starting from this level of knowledge it will be a rather simple task to actually solve complex systems of equations by matrix algebra (and matrix inversion). Without going into details, the principle is more or less to rewrite the example above such that $\mathbf{x=A^{-1}b}$ and then solve for the $\mathbf{x}$'s.

8.3 The determinant of a matrix

Determinants assign a scalar value (a single number) to a matrix. Determinants constitute important tools for economic analysis for many reasons. First, they allow us to check if there is indeed a solution to any given equation system (i.e, that we have enough linearly independent equations to solve the system). Second, determinants are important as they enable an evaluation of the sufficient conditions for an optimum. Third, determinants enable us to evaluate the effects of changes in exogenous parameters on the equilibrium values of our choice variables (so called comparative statics).

Determinants can only be calculated for square matrices (matrices with an equal number of rows and columns). The determinant of matrix $\mathbf{A}$ is written $det(\mathbf{A})$ or $|\mathbf{A}|$.

8.3.1 Calculating the determinant of a matrix

The smallest possible determinant is that of a 1×1 matrix $A = [a_{11}] : |A| = |a_{11}| = a_{11}$ (note that the brackets here are not related to absolute numbers.

Determinants can be both negative and positive). For a 2×2 matrix, the determinant is given by:

$$|\mathbf{A}| = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

In other words, we simply "draw a cross" in terms of multiplying the diagonal elements and subtracting the product of the second diagonal from the first. Using numbers, we have:

$$|\mathbf{A}| = \begin{bmatrix} 1 & 5 \\ 4 & 7 \end{bmatrix} = 7 - 20 = -13$$

When we have matrices of higher orders than 2x2, have to divide our calculation into several steps. Let us first show the general rule and then explain how we got there. The determinant of a 3x3 matrix is given by:

$$|\mathbf{A}| = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

$$|\mathbf{A}| = a_{11}[a_{22}a_{33} - a_{23}a_{32}] - a_{21}[a_{12}a_{33} - a_{13}a_{32}] + a_{31}[a_{12}a_{23} - a_{13}a_{22}]$$

The rule for deriving the determinant of any square matrix of an order higher than 2×2 is based on something called "Laplace expansion". It is based on the fact that we can partition large determinants into *sub determinants* or *minors* of smaller order than the original determinant. In the above example, we evaluated a 3×3 matrix. The determinant of that matrix, $|\mathbf{A}|$, can be divided into a number of minors. Each minor is acquired by deleting one row and one column in the "mother matrix". It is therefore said to be the minor of the element corresponding to that row and column. As such, the minor of element a_{11} is called $|\mathbf{M_{11}}|$ and given by:

$$|\mathbf{M_{11}}| = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

Similarly, we can define the minors of a_{21} and a_{31}:

$$|\mathbf{M_{21}}| = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

$$|\mathbf{M_{31}}| = \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

Note that we have defined the minors from the first column in $|A|$. We could, however, equally well have defined them from the second or third column (as an exercise, do this yourself!). The general representation of a minor is $|\mathbf{M_{ij}}|$, where i is the row number and j is the column number of the element to which the minor belongs.

In our example of a 3×3 matrix, the minors are 2×2 determinants, and can thus be evaluated. However, we still do not know how to use the resulting values. To calculate the determinant of $|\mathbf{A}|$, we need a rule. This rule is given by:

$$|\mathbf{A}| = \sum_{j=1}^{n} a_{ij}\mathbf{C_{ij}} = a_{ij}(-1)^{i+j}|\mathbf{M_{ij}}| \qquad \text{where} \quad j = 1, ..., n$$

$\mathbf{C_{ij}}$ is a so called *Cofactor*. It is derived by multiplying each minor by -1 raised to the sum of the row and the column number of $|\mathbf{M_{ij}}|$. In the example above, the cofactors are given by:

$$\mathbf{C_{11}} = (-1)^{1+1}|\mathbf{M_{11}}| = |\mathbf{M_{11}}| = \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

$$\mathbf{C_{21}} = (-1)^{2+1}|\mathbf{M_{21}}| = -|\mathbf{M_{21}}| = -\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

$$\mathbf{C_{31}} = (-1)^{3+1}|\mathbf{M_{31}}| = |\mathbf{M_{31}}| = \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

We are now ready to evaluate the determinant of a 3×3 matrix.

$$|\mathbf{A}| = a_{11}\mathbf{C_{11}} + a_{21}\mathbf{C_{21}} + a_{31}\mathbf{C_{31}}$$

$$|\mathbf{A}| = a_{11}|\mathbf{M_{11}}| - a_{21}|\mathbf{M_{21}}| + a_{31}|\mathbf{M_{31}}|$$

$$|\mathbf{A}| = a_{11}[a_{22}a_{33} - a_{23}a_{32}] - a_{21}[a_{12}a_{33} - a_{13}a_{32}] + a_{33}[a_{12}a_{23} - a_{13}a_{22}]$$

When we want to evaluate determinants of matrices of higher orders than 3×3, the calculation becomes a bit messy, and is most easily done by the help of a computer. It can however, also be done by hand. The rule to follow is the same as outlined above. However, in the case of larger matrices, we need to partition the original matrix more times in order to get down to 2×2 determinants. The basic rule is as follows:

$$|\mathbf{A}| = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

As before, we have that,

$$|\mathbf{A}| = a_{11}\mathbf{C_{11}} + a_{21}\mathbf{C_{21}} + a_{31}\mathbf{C_{31}} + a_{41}\mathbf{C_{41}}$$

$$|\mathbf{A}| = a_{11}(-1)^{1+1}|\mathbf{M_{11}}| + a_{21}(-1)^{2+1}|\mathbf{M_{21}}| + a_{31}(-1)^{3+1}|\mathbf{M_{31}}| + a_{41}(-1)^{4+1}|\mathbf{M_{41}}|$$

$$|\mathbf{A}| = a_{11}|\mathbf{M_{11}}| - a_{21}|\mathbf{M_{21}}| + a_{31}|\mathbf{M_{31}}| - a_{41}|\mathbf{M_{41}}|$$

However, the $|\mathbf{M_{ij}}|$ are no longer 2×2 matrices, but rather 3×3 matrices, e.g.,

$$|\mathbf{M_{11}}| = \begin{vmatrix} a_{22} & a_{23} & a_{24} \\ a_{32} & a_{33} & a_{34} \\ a_{42} & a_{43} & a_{44} \end{vmatrix}$$

which we have to evaluate in the same fashion as we did for the 3×3 matrix before. Thus, we have that:

$$|\mathbf{M_{11}}| = a_{22}(-1)^{1+1}[a_{33}a_{44} - a_{34}a_{43}] + a_{32}(-1)^{2+1}[a_{23}a_{44} - a_{24}a_{43}] + a_{42}(-1)^{3+1}[a_{23}a_{34} - a_{24}a_{33}]$$

$$|\mathbf{M_{11}}| = a_{22}[a_{33}a_{44} - a_{34}a_{43}] - a_{32}[a_{23}a_{44} - a_{24}a_{43}] + a_{42}[a_{23}a_{34} - a_{24}a_{33}]$$

Note that $|\mathbf{M_{11}}|$ forms a new matrix in which a_{22} is in the first row and first column, and that we therefore have to multiply its minor by $(-1)^{1+1}$ (see exercise MA03).

8.3.2 Solving equation systems by the use of Cramers rule

Suppose we want to solve the equation system below

$$\begin{aligned} Q_d - 2P &= 3 \\ Q_s - 4P &= 1 \\ Q_d - Q_s &= 0 \end{aligned}$$

we can write these equations in the now familiar matrix form $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}$ is the square coefficient matrix, $\mathbf{x}$ is the vector of variables with unknown values (Q_d, Q_s, P) and $\mathbf{b}$ is the vector of parameters with known values $(3, 1, 0)$.

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -4 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} Q_d \\ Q_s \\ P \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$$

Cramers rule tells us that, as long as $|\mathbf{A}| \neq 0$ we can find the solution to the equation system for one variable at the time by the use of the following formula:

$$x_i^* = \frac{|\mathbf{A_i}|}{|\mathbf{A}|}$$

where x_i is the i:th argument in the variable vector $\mathbf{x}$, $|\mathbf{A}|$ is the determinant of the 3x3 coefficient matrix and the determinant $|\mathbf{A_i}|$ is acquired by replacing the coefficients relating to x_i in matrix $\mathbf{A}$ by the parameter values in matrix $\mathbf{b}$. Hence, if we want to solve for x_1, we substitute b_1, b_2 and b_3 for a_{11}, a_{21} and a_{31} in $|\mathbf{A}|$ and evaluate the determinant.

$$|\mathbf{A_i}| = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{32} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} 3 & 0 & -2 \\ 1 & 1 & -4 \\ 0 & -1 & 0 \end{vmatrix}$$

Expanding $|\mathbf{A_i}|$ from the first column, we get the determinant value:

$$|\mathbf{A_i}| = 3 \times [1 \times 0 - (-4) \times (-1)] - 1 \times [0 \times 0 - (-2) \times (-1)] + 0 \times [0 \times (-4) - (-2) \times 1]$$

$$|\mathbf{A_i}| = 3 \times [-4] - 1 \times [-(-2)]$$

$$|\mathbf{A_i}| = -12 + 2 = -10$$

The determinant $|\mathbf{A}|$ is given by:

$$|\mathbf{A}| = 1 \times [0 - 4] + 1 \times [0 + 2] = -2$$

In our case, $x_1 = Q_d$. Dividing the solution value of $|\mathbf{A_i}|$ by $|\mathbf{A}|$, we have the solution value of $Q_d = -10/(-2) = 5$. If we repeat the procedure for Q_s, we see that the equilibrium condition indeed holds, $Q_s = Q_d = 5$. We leave it as an exercise to calculate the equilibrium price.

8.3.3 Exercises

1. Evaluate the determinants below

(a)

$$|\mathbf{A_1}| = \begin{vmatrix} 2 & 5 \\ 1 & 2 \end{vmatrix}$$

$$|\mathbf{A_2}| = \begin{vmatrix} -1 & 0 \\ 10 & 6 \end{vmatrix}$$

$$|\mathbf{A_3}| = \begin{vmatrix} 12 & -7 \\ 1 & 0 \end{vmatrix}$$

(b)

$$|\mathbf{B_1}| = \begin{vmatrix} 1 & 3 & -4 \\ 5 & 2 & -2 \\ -3 & -2 & 0 \end{vmatrix}$$

$$|\mathbf{B_2}| = \begin{vmatrix} 0 & 1 & 0 \\ 1 & -5 & 3 \\ 0 & 2 & 0 \end{vmatrix}$$

(c)

$$|\mathbf{C_1}| = \begin{vmatrix} 0 & 1 & 2 & 0 \\ 2 & 0 & -3 & 1 \\ 0 & 4 & 0 & 0 \\ -4 & 1 & 0 & 1 \end{vmatrix}$$

$$|\mathbf{C_2}| = \begin{vmatrix} 2 & 0 & -1 & 0 \\ 0 & -2 & 0 & 4 \\ 4 & 0 & 0 & 2 \\ 0 & -5 & 3 & 0 \end{vmatrix}$$

2. Calculate X, Y and Z by the use of Cramers rule

$$2X - 2Y + 2Z = 23$$
$$10X - 3Z = 0$$
$$2Y + 3Z = 33$$

3. Derive the expressions for $\frac{dx_1}{dM}$, $\frac{dx_2}{dM}$ and $\frac{d\lambda}{dM}$, by the use of Cramers rule on the equation system below

$$\begin{aligned} u_{11}\frac{dx_1}{dM} + u_{12}\frac{dx_2}{dM} - p_1\frac{d\lambda}{dM} &= 0 \\ u_{21}\frac{dx_1}{dM} + u_{22}\frac{dx_2}{dM} - p_2\frac{d\lambda}{dM} &= 0 \\ -p_1\frac{dx_1}{dM} - p_2\frac{dx_2}{dM} &= -1 \end{aligned}$$

Solutions 1.(a) $|\mathbf{A_1}| = -1$, $|\mathbf{A_2}| = -6$ and $|\mathbf{A_3}| = 7$.

1.(b) $|\mathbf{B_1}| = 30$ and $|\mathbf{B_2}| = 0$.

1.(c) $|\mathbf{C_1}| = 48$ and $|\mathbf{C_2}| = 104$

2. x=3 and y=1.5, z=10. To see this, note that:

$$|\mathbf{D}| = \begin{vmatrix} 2 & -2 & 2 \\ 10 & 0 & -3 \\ 0 & 2 & 3 \end{vmatrix} = 112$$

$$|\mathbf{D_x}| = \begin{vmatrix} 23 & -2 & 2 \\ 0 & 0 & -3 \\ 33 & 2 & 3 \end{vmatrix} = 336$$

$$|\mathbf{D_y}| = \begin{vmatrix} 2 & 23 & 2 \\ 10 & 0 & -3 \\ 0 & 33 & 3 \end{vmatrix} = 168$$

$$|\mathbf{D_z}| = \begin{vmatrix} 2 & -2 & 23 \\ 10 & 0 & 0 \\ 0 & 2 & 33 \end{vmatrix} = 1120$$

Hence, $x = 336/112 = 3$, $y = 168/112 = 1.5$ and $z = 1120/112 = 10$

3.

$$|\mathbf{D}| = 2p_1p_2u_{12} - p_2^2u_{11} - p_1^2u_{22}$$

$$\frac{dx_1}{dM} = \frac{p_2u_{12} - p_1u_{22}}{|\mathbf{D}|}$$

$$\frac{dx_2}{dM} = \frac{p_1u_{21} - p_2u_{11}}{|\mathbf{D}|}$$

$$\frac{d\lambda}{dM} = \frac{u_{21}u_{12} - u_{11}u_{22}}{|\mathbf{D}|}$$

A Math symbols and Notation

Summation Consider the European monetary union (EMU) and the total money supply. Total money supply is the sum of all member countries' money stock, i.e. $MS_{EMU} = MS_{Germany} + MS_{France} + MS_{Italy} + MS_{Spain} + MS_{Greek} + ...$ A much more convenient way to write this sum of money stock is to denote each country's money stock MS_i where $i = 1, ..., n$ (n is the total number of member countries) and write

$$MS_{EMU} = \sum_{i=1}^{n} MS_i \qquad \text{where} \quad i = 1, ..., n$$

In words, you would say 'the sum from $i = 1$ to n of MS_i'.

Greek letters Below you find a table of greek letters that are common to use in mathematic- and economic notation.

A	α	alpha
B	β	beta
Γ	γ	gamma
Δ	δ	delta
E	ϵ	epsilon
Z	ζ	zeta
H	η	eta
I	ι	iota
K	κ	kappa
Λ	λ	lambda
M	μ	mu
N	ν	nu
Ξ	ξ	xi
O	o	omicron
Π	π	pi
P	ρ	rho
Σ	σ	sigma
T	τ	tau
Υ	υ	upsilon
Φ	ϕ	phi
X	χ	chi
Ψ	ψ	psi
Ω	ω	omega